

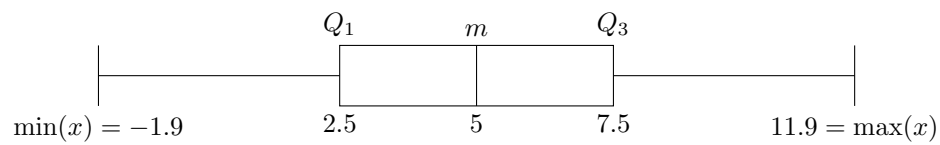
Statistics HW

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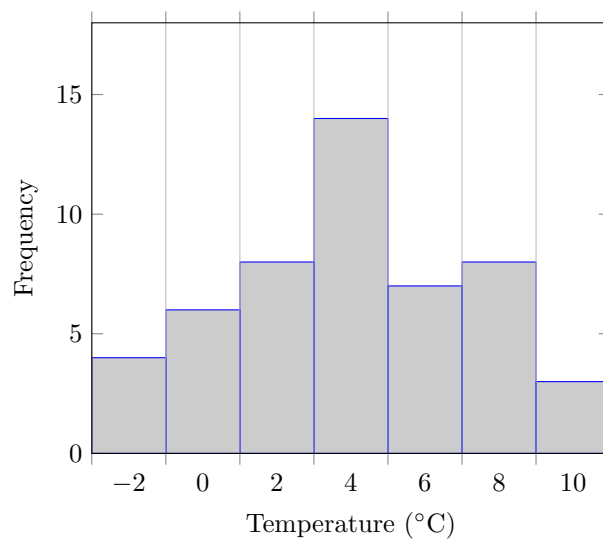
Question 1

(α)



(β)

Class	Frequency
$[-2, 0)$	4
$[0, 2)$	6
$[2, 4)$	8
$[4, 6)$	14
$[6, 8)$	7
$[8, 10)$	8
$[10, 12]$	3
Total	50



(γ)

Mean: $\frac{1}{50}(-4 + 6 + 24 + 70 + 49 + 72 + 33) = 5$

Variance: $\frac{1}{49}(4(6)^2 + 6(4)^2 + 8(2)^2 + 7(2)^2 + 8(4)^2 + 3(6)^2) = 10.939$

(δ)

- $x = 0$, which is the fifth datapoint, and is not the average of any two consecutive datapoints, thus $50p = (4, 5) \Rightarrow p = (0.08, 0.1)$
- $x = 6$, the only way to get 6 is for it be to the average of the 32nd and 33rd datapoint, thus $50p = 32 \Rightarrow p = 0.64$
- $x = 4$, which is the 19th, 20th and 21st datapoint, and is not the average of any two consecutive datapoints other than the fours, thus $50p = [18, 21) \Rightarrow p = [0.36, 0.42)$

Question 2

(α)

Both of these sampling statistics are normally distributed as a result of the Central Limit Theorem

$$S_{100} \approx \mathcal{N}(n\mu, n\sigma^2) = \mathcal{N}(200000, 16000000)$$

$$\bar{X} \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right) = \mathcal{N}(2000, 1600)$$

(β)

- $\mathbb{P}(Z < \frac{2049.2-2000}{40}) = \mathbb{P}(Z < 1.23) = 0.8907$
- $\mathbb{P}(Z > \frac{1980-2000}{40}) = 1 - \mathbb{P}(Z < -0.5) = \mathbb{P}(Z < 0.5) = 0.6915$

(γ)

$$\begin{aligned} \mathbb{P}\left(\frac{196000 - 200000}{4000} < Z < \frac{208000 - 200000}{4000}\right) &= \mathbb{P}(-1 < Z < 2) \\ &= \mathbb{P}(Z < 2) - (1 - \mathbb{P}(Z < 1)) = 0.9772 - (1 - 0.8413) = 0.8185 \end{aligned}$$

(δ)

$$Y_m \approx \mathcal{N}\left(2000, \frac{16000}{m}\right)$$

$$\mathbb{P}(\bar{Y} > 2030) = 1 - \mathbb{P}(\bar{Y} < 2030)$$

$$\Rightarrow \mathbb{P}(\bar{Y} < 2030) = 0.93319$$

$$\Rightarrow \mathbb{P}\left(Z < \frac{2030 - 2000}{\left(\frac{400}{\sqrt{m}}\right)}\right) = 0.93319$$

$$\Rightarrow \frac{30}{\left(\frac{400}{\sqrt{m}}\right)} = 1.5$$

$$\Rightarrow m = 400$$

Question 3

(α)

$$\bar{x} = \frac{1}{16} \sum_{i=1}^{16} x_i = 12$$

$$s = \sqrt{\frac{1}{15} \sum_{i=1}^{16} (x_i - \bar{x})^2} = 4$$

(β)

Since we are dealing with a sampling from a normal distribution, with both μ and σ^2 unknown, the only method at our disposal to estimate μ is the student distribution

$$(\bar{x} - t_{n-1; \alpha/2} \frac{s}{\sqrt{n}} < \mu < \bar{x} + t_{n-1; \alpha/2} \frac{s}{\sqrt{n}})$$

- $(12 - 1.753 < \mu < 12 + 1.753) = (10.247 < \mu < 13.753)$
- $(12 - 2.131 < \mu < 12 + 2.131) = (9.869 < \mu < 14.131)$

(γ)

- 9
 - Doesn't belong to the 88%
 - Doesn't belong to the 92%
 - Cannot be sure about the 98%
- 10
 - Doesn't belong to the 88%
 - Doesn't belong to the 92%
 - Belongs to the 98%
- 11
 - Cannot be sure about the 88%
 - Belongs to the 92%
 - Belongs to the 98%

(δ)

Now, when the standard deviation is known we can apply the formula

$$(\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}})$$

Thus the 95% confidence interval is

$$(12 - 1.96 < \mu < 12 + 1.96) = (10.355 < \mu < 13.645)$$

The length of a confidence interval is given by

$$l = 2z_{\alpha/2} \frac{\sigma}{\sqrt{m}}$$

Solving for m for an 95% confidence interval of at most length l_0 , we get

$$m \geq \left(\frac{15.68}{l_0} \right)$$

For $l_0 = 3$, and $l_0 = 2$, we get

$$m \geq 5.23, \quad m \geq 7.84$$

Then for $2 \leq l_0 \leq 3$, we have $6 \leq m \leq 7$

Question 4

(α)

$$\hat{p}_1 = \frac{3}{5}, \quad \hat{p}_2 = \frac{1}{2}$$

(β)

The $100(1 - \alpha)\%$ confidence interval for the percentage p of a characteristic is given by

$$p \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

- Then $(0.6 - 1.96\sqrt{\frac{(0.6)(0.4)}{50}} < p_1 < 0.6 + 1.96\sqrt{\frac{(0.6)(0.4)}{50}}) = (0.464 < p_1 < 0.736)$
- Also $(0.5 - 1.645\sqrt{\frac{0.5^2}{40}} < p_2 < 0.5 + 1.645\sqrt{\frac{0.5^2}{40}}) = (0.370 < p < 0.630)$

(γ)

$$\hat{p}_1 - \hat{p}_2 \pm z_{\alpha/2} \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}}$$

- $\left(0.1 - 2.576\sqrt{\frac{(0.6)(0.4)}{50} + \frac{0.5^2}{40}} < \hat{p}_1 - \hat{p}_2 < 0.1 + 2.576\sqrt{\frac{(0.6)(0.4)}{50} + \frac{0.5^2}{40}}\right)$
 $= (-0.171 < \hat{p}_1 - \hat{p}_2 < 0.371)$
- $\left(-0.1 - 1.96\sqrt{\frac{(0.6)(0.4)}{50} + \frac{0.5^2}{40}} < \hat{p}_2 - \hat{p}_1 < -0.1 + 1.96\sqrt{\frac{(0.6)(0.4)}{50} + \frac{0.5^2}{40}}\right)$
 $= (-0.306 < \hat{p}_2 - \hat{p}_1 < 0.106)$

(δ)

We recall the formula for the $100(1 - \alpha)\%$ confidence interval for the percentage p of a characteristic

$$p \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

Then for the characteristic q , we get

$$1 - p \pm z_{\alpha/2} \sqrt{\frac{(1 - \hat{p})\hat{p}}{n}}$$

We note that the length of C_p and C_q are the same since we are adding and subtracting the same term from both of the start points p and $q = 1 - p$. These are the same terms since we are dealing with the same dataset and each occurrence of the data is either p or q

Moreover, the two confidence intervals are both of equal distance away from $p = 0.5$, as a result of the property that $q = 1 - p$

Question 5

(A)

$$\begin{aligned} g(\theta; x_1, \dots, x_n) &= f_{X_1}(x_1)f_{X_2}(x_2) \dots f_{X_n}(x_n) \\ &= \theta^n \prod_{i=1}^n x_i^{-(1+\theta)} \end{aligned}$$

To maximise the likelihood, it suffices to do it for the log of the function

$$\begin{aligned} h(\theta) &= \ln(g(\theta; X_1 \dots X_n)) \\ &= n \ln \theta - (1 + \theta) \sum_{i=1}^n \ln X_i \\ h'(\theta) &= \frac{n}{\theta} - \sum_{i=1}^n \ln X_i \end{aligned}$$

When $h'(\hat{\theta}) = 0$, we have

$$\frac{n}{\bar{\theta}} = \sum_{i=1}^n \ln X_i$$

$$\Rightarrow \hat{\theta} = \frac{n}{\sum_{i=1}^n Y_i}$$

To check whether this is a maximum or not we look at the second derivative of $h(\theta)$

$$h''(\theta) = -\frac{n}{\theta^2}$$

$$h''(\hat{\theta}) = -\frac{n}{\hat{\theta}^2}$$

Since $n > 0$ and $\hat{\theta}^2 > 0$, we have that this is indeed a maximum and thus

$$\text{MLE}(\theta) = \hat{\theta} = \frac{n}{\sum_{i=1}^n Y_i} = \frac{1}{\bar{Y}}$$

(B)

•

$$\bar{x} = \frac{x_1 + \cdots + x_n}{n}, \quad \bar{y} = \frac{y_1 + \cdots + y_m}{m}$$

Let $z = \{x_1, \dots, x_n, y_1, \dots, y_m\}$

$$\bar{z} = \frac{x_1 + \cdots + x_n + y_1 + \cdots + y_m}{n + m}$$

$$= \frac{n\bar{x} + m\bar{y}}{n + m}$$

•

$$s_x^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2, \quad s_y^2 = \frac{1}{m-1} \sum_{i=1}^m (y_i - \bar{y})^2$$

Then

$$s^2 = \frac{1}{n+m-1} \sum_{i=1}^{n+m} (z_i - \bar{z})^2$$

$$= \frac{1}{n+m-1} \left(\sum_{i=1}^n (x_i - \bar{z})^2 + \sum_{i=1}^m (y_i - \bar{z})^2 \right)$$

$$= \frac{1}{n+m-1} \left(\sum_{i=1}^n (x_i - \bar{x} + \bar{x} - \bar{z})^2 + \sum_{i=1}^m (y_i - \bar{y} + \bar{y} - \bar{z})^2 \right)$$

$$= \frac{1}{n+m-1} \left(\sum_{i=1}^n (x_i - \bar{x})^2 + 2(\bar{x} - \bar{z}) \sum_{i=1}^n (x_i - \bar{x}) + n(\bar{x} - \bar{z})^2 + \sum_{i=1}^m (y_i - \bar{y})^2 + 2(\bar{y} - \bar{z}) \sum_{i=1}^m (y_i - \bar{y}) + m(\bar{y} - \bar{z})^2 \right)$$

$\left(\sum_{i=1}^n (x_i - \bar{x}) = 0 \right)$ Similarly for y

$$= \frac{1}{n+m-1} \left((n-1)s_x^2 + (m-1)s_y^2 + n(\bar{x} - \bar{z})^2 + m(\bar{y} - \bar{z})^2 \right)$$

$$= \frac{1}{n+m-1} \left((n-1)s_x^2 + (m-1)s_y^2 + \frac{nm}{n+m} (\bar{x} - \bar{y})^2 \right)$$

(C)

$$f(t) = \sum_{i=1}^n |x_i - t|, \quad t \in \mathbb{R}$$

With k being the index where the x_i 's become greater than t , we can rewrite this as

$$f(t) = \sum_{i=1}^k (t - x_i) + \sum_{i=k+1}^n (x_i - t)$$

Then

$$f'(t) = k - (n - k) = 2k - n$$

Now let $m = \frac{n+1}{2}$. Since n is odd, x_m is the median of the dataset.

If $t < x_m$, then $k < m$, so

$$f'(t) = 2k - n < 0$$

and hence $f(t)$ is decreasing.

If $t > x_m$, then $k \geq m$, so

$$f'(t) = 2k - n > 0$$

and hence $f(t)$ is increasing.

Therefore $f(t)$ decreases for $t < x_m$ and increases for $t > x_m$, so the minimum occurs at

$$t^* = x_m = x_{\frac{n+1}{2}}.$$

Thus the value of t that minimizes $f(t)$ is the median of the observations.

(D)

- Let $z = \{1, 2, \dots, n\}$

$$\text{Then } \bar{z} = \frac{1+2+\dots+n}{n} = \frac{n+1}{2}$$

We also know that the variance is given by $\frac{1}{n-1}(\sum i^2) - n(\bar{x})^2$

$$\text{Thus } s^2 = \frac{1}{n-1} \left(\frac{n(n+1)(2n+1)}{6} - n \left(\frac{n+1}{2} \right)^2 \right) = \frac{n(n+1)}{12}$$

- Then from the Exercise 1.3 we have the sample mean of the new set being added by the constant alpha $\frac{n+1+2\alpha}{2}$ and the sample variance being the same $\frac{n(n+1)}{12}$